



Ministry of  
Education  
Government of India



MoE's  
INNOVATION CELL  
(GOVERNMENT OF INDIA)



SMART INDIA  
HACKATHON  
2026

OFFICIAL PLAYBOOK • STUDENT EDITION

# SMART INDIA HACKATHON 2026 PLAYBOOK

A complete strategy guide to selecting the right Problem Statement, generating winning Open Innovation ideas, and building a pitch deck that gets you to the Grand Finale.



## SMART INDIA HACKATHON 2026

### #SmartIndiaHackathon

World's Biggest Open Innovation Model

COVERS

PS Selection  
Framework

INCLUDES

Top 10 Open Innovation  
Ideas

BONUS

Best PPT / Pitch Tools  
2026

# What's Inside This Playbook

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This playbook is built specifically for Smart India Hackathon 2026 teams — from problem statement shortlisting all the way to the final pitch. Use it as a working reference through your internal hackathon and national submission.

|                                                           |           |
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**How to use this playbook:** Read Sections 2-3 before you touch the problem statement list. Read Section 5 only for inspiration — SIH judges reward original thinking, so treat these as starting points to customise, not copy.

Smart India Hackathon is India's largest nationwide innovation initiative, organised by the Ministry of Education's Innovation Cell (MIC) and AICTE, inviting students to build real, deployable solutions for problems submitted by government ministries, PSUs, and industry partners.

|                            |                                     |                               |                                |
|----------------------------|-------------------------------------|-------------------------------|--------------------------------|
| <b>6</b><br>MEMBERS / TEAM | <b>36 hrs</b><br>GRAND FINALE BUILD | <b>2</b><br>EDITIONS: SW & HW | <b>1</b><br>FEMALE MEMBER MIN. |
|----------------------------|-------------------------------------|-------------------------------|--------------------------------|

### The Journey, Stage by Stage

| STAGE                      | WHAT HAPPENS                                                              | WHAT GETS YOU ELIMINATED                                  |
|----------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>Team Formation</b>      | Form a 6-member cross-functional team under a faculty SPOC                | Skill overlap with no domain/design/business balance      |
| <b>PS Selection</b>        | Browse published software & hardware problem statements and shortlist 3-5 | Picking on "coolness" instead of team-fit and feasibility |
| <b>Internal Hackathon</b>  | Compete within your own institution to earn SPOC nomination               | Presenting only slides, no working demo                   |
| <b>National Submission</b> | SPOC uploads your idea PPT and demo video to the national portal          | Weak problem understanding, vague impact metrics          |
| <b>Grand Finale</b>        | 36-hour non-stop build at a nodal centre, judged live                     | Prototype that doesn't run, poor time management          |

### Sectors Ministries Are Prioritising in 2026

- ✓ Healthcare — AI diagnostics, telemedicine, patient records
- ✓ Agriculture & Food — smart farming, crop & supply chain
- ✓ Education — personalised learning, skilling, assessment
- ✓ Smart Cities / IoT — automation, traffic, utilities
- ✓ Sustainability — carbon tracking, renewables, waste mgmt.
- ✓ Cybersecurity — digital identity, secure transactions
- ✓ Disaster Management & Emergency Response
- ✓ Tourism, Transportation & Mobility

**Why this matters:** Judges in 2026 explicitly expect meaningful AI integration — not AI for the sake of AI. A well-scoped, boring-sounding idea with a working model will consistently beat a flashy idea with mock screens.

# How to Choose the Right Problem Statement

Problem statement selection is the single biggest predictor of SIH success. Most teams lose not because of bad execution, but because they picked a PS that was either too competitive, too vague, or mismatched with their skills.

## The 5-Step Selection Method

### Step 1 — Audit Your Team's Real Skills First

Before opening the PS list, list out your team's actual strengths: who codes what, who can design, who understands the domain (health, agri, finance), and who can present. Choose a PS your team can execute — not one that merely sounds impressive to judges. Feasibility is scored explicitly at every stage.

### Step 2 — Read Every Shortlisted PS at Least 10 Times

Most teams fail because they solve the wrong problem. Read the department's exact wording, the expected outcome, and any constraints mentioned. Underline the actual pain point — not the feature you assume they want.

### Step 3 — Filter for "Impactful but Less Competitive"

Popular, headline-sounding problem statements (e.g. generic chatbots, generic dashboards) attract hundreds of teams nationally. Look one level deeper — a well-defined statement with real but under-served pain (e.g. a niche compliance workflow, a rural logistics gap) gives you far less competition for the same visibility.

### Step 4 — Run a Root-Cause Check

For each shortlisted PS, ask "why does this problem exist today?" three times. If your solution only treats a symptom (e.g. "send SMS reminders") rather than the root cause (e.g. "no single source of verified data"), keep digging before committing.

### Step 5 — Validate With a Feasibility Sketch

Before finalising, sketch (on paper) what a working demo would look like in 36 hours. If you cannot picture an end-to-end demo — even a rough one — the PS is too broad. Narrow the scope until a demo becomes imaginable.

**Common mistake:** Choosing a PS because "it sounds like our final year project." Judges can tell when a solution is retrofitted onto a problem statement rather than built for it — problem understanding is scored as a separate criterion.

## Software vs Hardware Edition — Which to Pick?

| FACTOR                | SOFTWARE EDITION               | HARDWARE EDITION                              |
|-----------------------|--------------------------------|-----------------------------------------------|
| Best for teams with   | Strong app/web/AI-ML skills    | Electronics, IoT, embedded skills             |
| Demo risk             | Lower — cloud/local demo       | Higher — physical failure risk on stage       |
| Differentiation       | Harder (more competition)      | Easier if you have real hardware access       |
| Typical judging focus | UX, scalability, data handling | Build quality, cost-effectiveness, robustness |

# The PS Shortlisting Scorecard

Score every candidate problem statement out of 5 on each factor below. Anything scoring under 15/25 total should be dropped from your shortlist, regardless of how "exciting" it sounds.

| CRITERIA                     | 1 (WEAK)                               | 3 (AVERAGE)                     | 5 (STRONG)                              |
|------------------------------|----------------------------------------|---------------------------------|-----------------------------------------|
| <b>Team Skill Fit</b>        | Requires tech nobody on the team knows | Partial overlap, needs learning | Matches 2+ members' core strengths      |
| <b>Clarity of Ask</b>        | Vague, open to many interpretations    | Somewhat defined outcome        | Clear department, clear expected output |
| <b>Feasibility in 36 hrs</b> | Needs data/hardware you can't access   | Achievable with workarounds     | End-to-end demo is realistic            |
| <b>Competition Level</b>     | Generic, high-visibility topic         | Moderately common               | Niche but genuinely needed              |
| <b>Real-World Impact</b>     | Impact hard to measure/explain         | Some measurable benefit         | Clear beneficiary + measurable metric   |

**Pro tip:** Score at least 5 problem statements before your internal hackathon, not just your favourite one. Teams that shortlist a backup PS rarely lose momentum if their first choice gets contested internally.

## Red Flags to Drop a PS Immediately

- ✓ The department's expected outcome is a one-line vague sentence with no context
- ✓ Your solution idea already exists as a mature product with no clear gap to improve
- ✓ Building even a partial demo needs licensed data, paid APIs, or hardware you cannot source
- ✓ No one on your team can explain the domain problem in plain language without notes
- ✓ The PS description contradicts itself or has been reused for 3+ hackathon cycles with no fresh angle

## Green Flags — Signals of a Strong PS

- ✓ The ministry/organisation names a specific current gap (not a generic ask)
- ✓ The scope can realistically produce a working, clickable/functional prototype in 36 hours
- ✓ Your team already has partial domain knowledge or personal exposure to the problem
- ✓ The expected outcome allows room for a creative, non-obvious technical approach

The Open Innovation theme has no fixed ministry brief — teams propose their own problem and solution. This freedom is exactly why most Open Innovation entries fail: without a real department scoping the ask, teams drift into generic, "nice to have" ideas. The strongest Open Innovation submissions borrow the discipline of a ministry PS even though none was assigned.

## What Consistently Wins in This Theme

- ✓ A narrow, well-defined user group (not "everyone in India")
- ✓ A believable path to real deployment (cost, scale, adoption)
- ✓ A problem the team has personally observed or experienced
- ✓ Clear before/after metrics: time saved, cost reduced, lives improved
- ✓ Working offline-first design for low-connectivity regions
- ✓ A demo that runs live, even if the dataset is small
- ✓ Meaningful use of AI/ML — prediction, vision, or NLP — not decoration
- ✓ Original framing — a familiar problem solved in an unfamiliar way

**Judging lens for Open Innovation:** Novelty of approach, technical depth, feasibility of the working prototype, scalability, cost-effectiveness, and the size/clarity of real-world impact. Every idea in Section 5 is scored against these six lenses.

## Trending Sub-Themes for 2026

Based on ministry priorities and where student innovation has shown strongest traction recently, five sub-themes stand out for Open Innovation in 2026: AI-assisted public health screening, climate-resilient agriculture, disaster early-warning systems, accessible digital learning, and citizen-facing governance transparency tools.

# Top 10 Open Innovation Ideas

Use these as starting frameworks, not final pitches. Add your own domain twist, local data, and a differentiated technical approach before you present — originality is scored, and judges see repeated ideas every season.

## 1 Offline AI Health Screening Companion

Healthcare AI/ML Rural

A smartphone app that screens for common conditions (anemia, diabetes risk, eye disorders, skin conditions) using camera input and basic vital signs, running fully offline with periodic sync — built for regions with poor connectivity and a shortage of diagnostic staff.

## 2 Hyperlocal Crop Advisory & Disease Detection

Agriculture Computer Vision IoT

A vision-based tool that identifies crop disease and nutrient deficiency from a phone photo, cross-referenced with hyperlocal soil and weather data, giving farmers a specific, low-cost remedy instead of a generic pesticide recommendation.

## 3 AI-Powered Disaster Early-Warning & Evacuation Router

Disaster Mgmt Predictive AI Maps

Combines weather, terrain, and historical flood/landslide data to predict risk zones, then generates dynamic, capacity-aware evacuation routes for local authorities — with an SMS fallback layer for areas without smartphones.

## 4 Adaptive Micro-Learning for Regional Language Learners

EdTech NLP Accessibility

An adaptive learning engine that restructures school content into bite-sized, regional-language micro-lessons based on a student's live performance, targeting the learning gap in low-resource schools rather than another generic quiz app.

## 5 Citizen Grievance-to-Resolution Transparency Tracker

Governance NLP Civic Tech

Automatically classifies and routes citizen complaints (roads, water, sanitation) to the correct department using NLP, then publishes a transparent public resolution timeline — directly tackling the trust gap in existing grievance portals.

# Top 10 Open Innovation Ideas

6

## Smart Waste Segregation & Route Optimisation for Municipalities

Sustainability IoT Smart Cities

Low-cost sensor-based bins that detect fill level and waste type, feeding a route-optimisation engine so municipal trucks collect only when and where needed — cutting fuel cost and overflow complaints simultaneously.

7

## AI Copilot for MSME Compliance & Documentation

Fintech NLP MSME

A conversational assistant that walks small business owners through GST filing, licensing, and compliance paperwork in their own language, flagging errors before submission — targeting the compliance drop-off that keeps MSMEs informal.

8

## Wearable Fall & Distress Detection for Elderly Care

Healthcare Hardware/IoT Safety

A low-cost wearable that detects falls or prolonged inactivity using an accelerometer and lightweight on-device ML, auto-alerting a family member or nearest health worker — designed for elderly living alone in semi-urban and rural India.

9

## Water Quality & Leak Detection Network for Small Towns

Sustainability IoT Public Health

A mesh of low-cost sensors monitoring contamination levels and pipeline leaks in municipal water supply, giving local bodies real-time alerts instead of relying on manual, delayed testing — directly reducing waterborne disease risk.

10

## AI-Assisted Mental Wellness Companion for Students

Healthcare NLP EdTech

A stigma-free, anonymous chat companion for students that detects early signs of stress or burnout through conversation patterns and nudges them toward campus counsellors — built with strict privacy-by-design rather than as a data-collection tool.

**Before you commit to any idea above:** Search the SIH portal's archive and recent GitHub/hackathon repositories for similar past submissions. Judges specifically check for incremental improvement over prior years — add a genuinely new technical angle, dataset, or deployment model.



A polished slide with mockups will not beat a rough but working prototype. Structure your 36 hours (and the weeks before) around a demo-first build, not a design-first one.

### Recommended Team Role Split (6 members)

| ROLE                       | RESPONSIBILITY                                                                 |
|----------------------------|--------------------------------------------------------------------------------|
| Tech Lead                  | Owns architecture decisions, core backend/ML logic                             |
| Frontend/UX Owner          | Builds the interface, keeps the demo flow simple and clear                     |
| Domain Researcher          | Validates the problem, gathers real data/stats, talks to end-users if possible |
| Integration Engineer       | Connects APIs, hardware, or data pipelines end-to-end                          |
| QA & Demo Owner            | Stress-tests the demo path, prepares a fallback for live failures              |
| Pitch & Documentation Lead | Builds the PPT, writes the narrative, rehearses the pitch                      |

### 36-Hour Build Timeline (Suggested)

- ✓ **Hours 0-4:** Finalise architecture, lock scope, split tasks — do not code before this
- ✓ **Hours 4-20:** Build the core end-to-end path first (even ugly), before polishing anything
- ✓ **Hours 20-28:** Integrate all pieces, connect data, fix breaking bugs
- ✓ **Hours 28-32:** Polish UI/UX only after the core flow works reliably
- ✓ **Hours 32-36:** Rehearse the pitch, prepare backup demo (video/screenshots), sleep in shifts

**Judges' most common complaint:** "The prototype didn't work live." Always carry a recorded backup demo video on a local device, not just a live version dependent on venue Wi-Fi.

Your national-round submission is judged heavily on the PPT before anyone sees your code. These tools help you move fast without sacrificing a clean, judge-friendly structure.

**Gamma** FASTEST AI-GENERATED DECKS

Generates a structured, web-native deck from a prompt or outline in under a minute. Great for a first draft of your SIH PPT structure; export to PowerPoint before final submission since native export can lose some formatting.

**Canva (AI features)** VISUAL POLISH & TEMPLATES

Best asset library and design templates for making a technical deck look professional fast. Ideal for adding diagrams, icons, and consistent branding across your SIH submission.

**Microsoft PowerPoint + Copilot** NATIVE SIH-FORMAT COMPLIANCE

Since SIH mandates a specific PPT template, building directly in PowerPoint with Copilot assistance for content drafting keeps formatting fully compliant while saving drafting time.

**Beautiful.ai** AUTOMATIC DESIGN ENFORCEMENT

Auto-adjusts layout and spacing as you edit, so technical teams without a designer still produce a visually consistent deck — useful when your pitch lead has limited design time.

**Plus AI** POWERPOINT/GOOGLE SLIDES INTEGRATION

Works as an add-on inside PowerPoint or Google Slides, generating and refining slide content in your existing file rather than a separate platform — useful for teams already committed to the official template.

**Napkin AI** TURNING TEXT INTO DIAGRAMS

Converts written explanations of your architecture or workflow into clean visual diagrams automatically — useful for the "system architecture" slide judges always ask about.

**Reminder:** Whichever tool you use, the final file must follow the official SIH PPT template and format shared on the SIH portal — download and cross-check the mandated slide order before final upload.

Judges spend very little time per slide during screening. Every slide must earn its place — structure matters as much as content.

| # | SLIDE                     | WHAT IT MUST ANSWER                                              |
|---|---------------------------|------------------------------------------------------------------|
| 1 | Title & Team              | PS number, title, team name, institution                         |
| 2 | Problem Understanding     | What is the real problem, who suffers, how big is it (with data) |
| 3 | Proposed Solution         | What you're building, in one clear sentence                      |
| 4 | Innovation & Uniqueness   | What's different from existing solutions                         |
| 5 | Technical Approach        | Architecture diagram, tech stack, data flow                      |
| 6 | Feasibility & Prototype   | What already works, what's demoable today                        |
| 7 | Impact & Scalability      | Who benefits, measurable outcome, path to scale                  |
| 8 | Business/Deployment Model | Cost, sustainability, who adopts it and how                      |
| 9 | Team & Roles              | Why this team can execute this specific PS                       |

**Storytelling tip:** Open with a real (or realistic) beneficiary scenario — a farmer, a patient, a citizen — before diving into the technical solution. Judges retain narratives better than feature lists, and it immediately proves problem understanding.

# Final Checklist Before Submission

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## Problem Statement & Idea

- ✓ Scored at least 5 candidate PS on the scorecard in Section 3
- ✓ Root cause identified — not just a surface-level symptom
- ✓ Checked for duplicate/near-duplicate past submissions
- ✓ Idea has a clear, named beneficiary and measurable impact

## Prototype

- ✓ Core end-to-end flow works, even if UI is unpolished
- ✓ Backup demo video recorded locally (no dependency on venue Wi-Fi)
- ✓ Tested on a device/browser different from the one used to build it

## Presentation

- ✓ Follows the official SIH PPT template exactly (slide count, order, format)
- ✓ Every slide answers one clear question — no filler slides
- ✓ Rehearsed as a team at least twice against a timer
- ✓ Video demonstration recorded, clear audio, under time limit

## Team Readiness

- ✓ Every member can explain the problem statement without notes
- ✓ Roles for the 36-hour build are pre-assigned (see Section 6)
- ✓ Contingency plan exists if one member is unavailable during finale

**You're ready when:** Any single team member — not just the presenter — can walk a stranger through the problem, the solution, and the demo in under two minutes. That is the real test SIH judges apply.